DataFingers

NETWORK TRAFFIC ANALYSIS REPORT

Task Title: Network Traffic Analysis – Wireshark Live Capture

Name: Favour Chiemela

Tool Used: Wireshark

Capture Duration: Approximately 30–35 Minutes

Total Packets Captured: 88,959 Packets

1. Introduction:

As part of this task, I used Wireshark to capture and analyze live network traffic on my system. The aim was to observe how devices and applications communicate over a network and identify any interesting patterns that an investigator might notice during an investigation.

The traffic was captured while carrying out normal internet activities such as browsing websites and accessing online services. A total of 88,959 packets were captured during the session.

2. Findings and Observations:

Finding 1: DNS Requests to Online Services:

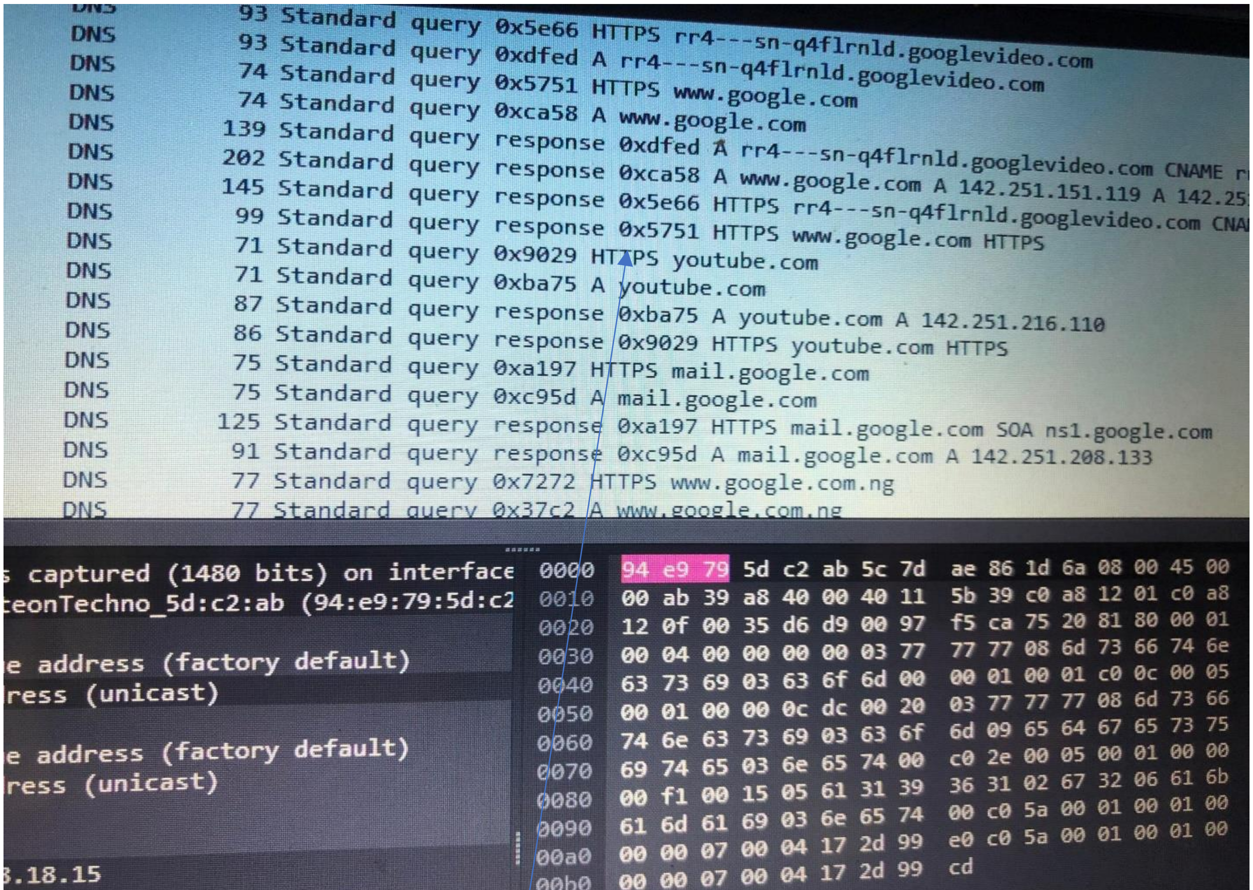
During the analysis, I observed approximately 320 DNS packets. DNS is responsible for translating website names into IP addresses so that devices can communicate with them.

Some of the domains observed included:

- www.youtube.com
- mail.google.com

This shows that before connecting to a website or service, the system first requested the address information needed to reach that service.

This helped to understand which online services a user is trying to access. And this shows that when later communications are encrypted, DNS requests can still provide useful clues about user activity.

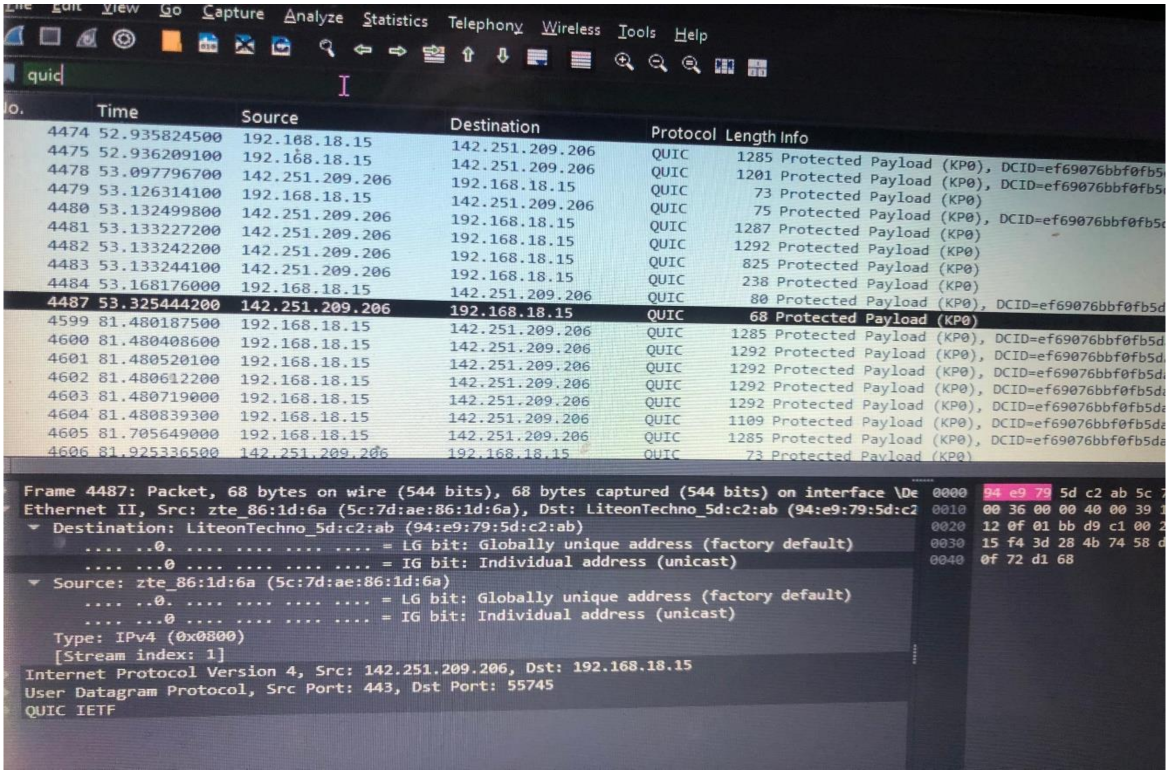


As an investigator, I checked the DNS list for unfamiliar or suspicious domains. If a device were infected with malware, we would see hidden DNS queries to strange, random website names that the user never typed. In this capture, every DNS query matched a safe website I actually visited.

Finding 2: High Volume of QUIC Traffic:

The capture contained approximately 37,407 QUIC packets, making it one of the most common protocols observed during the session.

QUIC is a modern protocol used by many websites and applications to make internet communication faster and smoother. During the analysis, I observed QUIC Initial packets, Handshake packets, and Protected Payload packets. Repeated communication with the same external addresses was also observed.

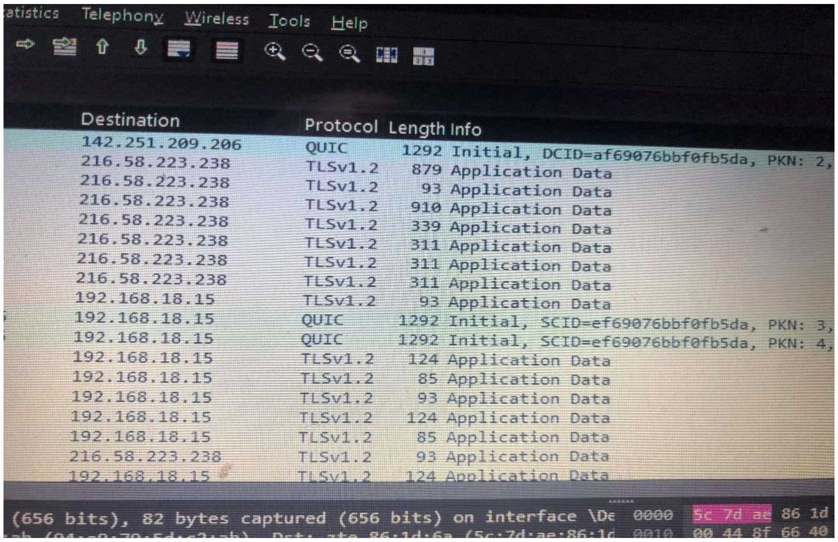


QUIC packets made up over 40% of the traffic because I was actively using Google services like YouTube and Gmail during the capture. QUIC is Google's custom protocol designed to speed up video streaming and web loading. This volume is completely normal and matches my actual activity.

Finding 3: Encrypted TLS Communication:

TLS 1.2 and TLS 1.3 traffic were observed throughout the capture. Many of the packets appeared as Application Data, which means the information being exchanged was protected through encryption.

Although the actual contents of the communication could not be viewed, it was clear that secure communication was taking place between the device and online services and that traffic patterns can still be analyzed.

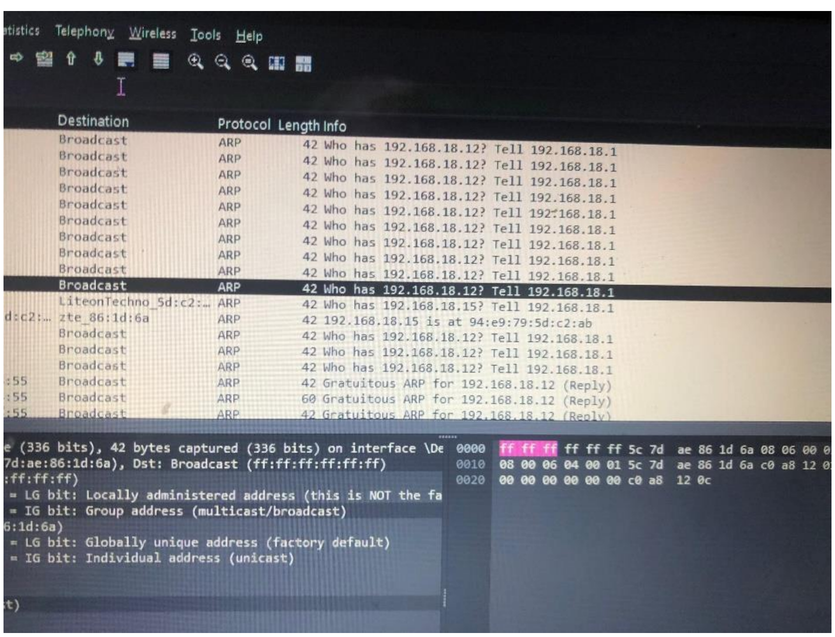


As indicated, the presence of TLS confirms that communication was completely encrypted. This proves that even if an attacker intercepted these specific packets on the network, they would see nothing but unreadable information.

Finding 4: ARP Activity on the Local Network:

Approximately 129 ARP packets were observed during the capture.

Messages such as "Who has..." and "Is at..." were present. These messages help devices find and identify each other on the same network before communication begins.



3. Overall Analysis:

The network traffic captured during this exercise showed normal user activity and communication with online services. The analysis revealed frequent DNS requests, extensive use of encrypted communication through TLS, a significant amount of QUIC traffic, and normal ARP activity within the local network.

No obvious signs of malicious activity, unauthorized access attempts, or network attacks were observed during the capture period.

4. Conclusion:

This exercise helped me understand how devices communicate across a network and how Wireshark can be used to observe that communication. The analysis showed that many activities take place in the background whenever a user visits a website or uses an online service. It also demonstrated the importance of encryption in protecting information while it travels across the internet. Overall, this task provided practical experience in network traffic analysis and improved my understanding of how network communication works.